LEAGUE OF LEGENDS

TRABALHO FINAL
FUNDAMENTOS DE BANCO DE DADOS

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DATASET

DATASET

A base de dados conta com 119 colunas e 21147 linhas contendo informações sobre partidas competitvas de League of Legends entre 2017 e 2024.
O dataset contém informações sobre campeonatos, equipes, jogadores e campeões.

Team1_region	Team1_Baron	Team1_Dra	Team1_Turts	Team1_ban1	Team1_ban2	Team1_ban3	Team1_ban4	Team1_ban5
Team1_player1_name	Team1_player1_pick	Team1_player1_K	Team1_player1_D	Team1_player1_A	Team1_player1_CS	Team1_player1_gold	Team1_player1_damage	Team1_player1_tanking
Team1_player2_name	Team1_player2_pick	Team1_player2_K	Team1_player2_D	Team1_player2_A	Team1_player2_CS	Team1_player2_gold	Team1_player2_damage	Team1_player2_tanking
Team1_player3_name	Team1_player3_pick	Team1_player3_K	Team1_player3_D	Team1_player3_A	Team1_player3_CS	Team1_player3_gold	Team1_player3_damage	Team1_player3_tanking
Team1_player4_name	Team1_player4_pick	Team1_player4_K	Team1_player4_D	Team1_player4_A	Team1_player4_CS	Team1_player4_gold	Team1_player4_damage	Team1_player4_tanking
Team1_player5_name	Team1_player5_pick	Team1_player5_K	Team1_player5_D	Team1_player5_A	Team1_player5_CS	Team1_player5_gold	Team1_player5_damage	Team1_player5_tanking
Team2_region	Team2_Baron	Team2_Dra	Team2_Turts	Team2_ban1	Team2_ban2	Team2_ban3	Team2_ban4	Team2_ban5
Team2_player1_name	Team2_player1_pick	Team2_player1_K	Team2_player1_D	Team2_player1_A	Team2_player1_CS	Team2_player1_gold	Team2_player1_damage	Team2_player1_tanking
Team2_player2_name	Team2_player2_pick	Team2_player2_K	Team2_player2_D	Team2_player2_A	Team2_player2_CS	Team2_player2_gold	Team2_player2_damage	Team2_player2_tanking
Team2_player3_name	Team2_player3_pick	Team2_player3_K	Team2_player3_D	Team2_player3_A	Team2_player3_CS	Team2_player3_gold	Team2_player3_damage	Team2_player3_tanking
Team2_player4_name	Team2_player4_pick	Team2_player4_K	Team2_player4_D	Team2_player4_A	Team2_player4_CS	Team2_player4_gold	Team2_player4_damage	Team2_player4_tanking
Team2_player5_name	Team2_player5_pick	Team2_player5_K	Team2_player5_D	Team2_player5_A	Team2_player5_CS	Team2_player5_gold	Team2_player5_damage	Team2_player5_tanking
MatchID	matchType	gameset	MatchDate	Duration	win	Team1	Team2	

NORMALIZAÇÃO

1º FORMA NORMAL

NORMALIZAÇÃO (1FN)

"Uma entidade relacional satisfaz o requisito da primeira forma normal se cada instância da entidade contiver apenas um valor, mas nunca vários atributos repetidos."

"Atributos repetitivos, geralmente chamados de grupo repetitivo, são atributos diferentes que são inerentemente iguais."

IBM NORMALIZAÇÃO NO DESIGN DE BANCO DE DADOS

Usaremos como chave primária **matchID** e **gameset**, pois identificam um jogo único

Repetição de uma Entidade "**Ban**"

Team1_Ban1	Team1_Ban2	Team1_Ban3	Team1_Ban4	Team1_Ban5

Team2_Ban1	Team2_Ban2	Team2_Ban3	Team2_Ban4	Team2_Ban5

Criamos a tabela **Ban**

matchID	**gameset**	**teamName**	**banOrder**	champion

NORMALIZAÇÃO (1FN)

"Uma entidade relacional satisfaz o requisito da primeira forma normal se cada instância da entidade contiver apenas um valor, mas nunca vários atributos repetidos."

"Atributos repetitivos, geralmente chamados de grupo repetitivo, são atributos diferentes que são inerentemente iguais."

IBM NORMALIZAÇÃO NO DESIGN DE BANCO DE DADOS

Team1_Name	Team1_Region	Team1_Baron	Team1_Dragon	Team1_Turrets

Team2_Name	Team2_Region	Team2_Baron	Team2_Dragon	Team2_Turrets

Criamos a tabela **Game_Team**

matchID	**gameset**	**teamName**	teamRegion	baron	dragon	turrets

NORMALIZAÇÃO (1FN)

“Uma entidade relacional satisfaz o requisito da primeira forma normal se cada instância da entidade contiver apenas um valor, mas nunca vários atributos repetidos.”

“Atributos repetitivos, geralmente chamados de grupo repetitivo, são atributos diferentes que são inerentemente iguais.”

IBM NORMALIZAÇÃO NO DESIGN DE BANCO DE DADOS

Repetição de uma Entidade “**Player_Game_Stats**“.

T1P1_Name	T1P1_Pick	T1P1_Kills	T1P1_Deaths	T1P1_Assists	T1P1_Farm	T1P1_Gold	T1P1_Damage	T1P1_Tanking

(...)

T2P5_Name	T2P5_Pick	T2P5_Kills	T2P5_Deaths	T2P5_Assists	T2P5_Farm	T2P5_Gold	T2P5_Damage	T2P5_Tanking

Criamos a tabela **Player_Game_Stats**

<u>matchID</u>	**<u>gameset</u>**	**playerName**	team	Pick	Kills	Deaths	Assists	Farm	Gold	Damage	Tanking

APÓS 1FN

Ban

<u>**matchID**</u>	<u>**gameset**</u>	**teamName**	**banOrder**	champion

Game_Team

<u>**matchID**</u>	<u>**gameset**</u>	**teamName**	teamRegion	baron	dragon	turrets

Player_Game_Stats

<u>**matchID**</u>	<u>**gameset**</u>	**playerName**	team	Pick	Kills	Deaths	Assists	Farm	Gold	Damage	Tanking

Match

matchID	**gameset**	matchType	matchDate	matchDuration	winner

2º FORMA NORMAL

NORMALIZAÇÃO (2FN)

"Uma entidade está na segunda forma normal se cada atributo que não estiver na chave primária fornecer um fato que dependa da chave inteira."

BAN

<u>**matchID**</u>	<u>**gameset**</u>	**teamName**	**banOrder**	champion

NORMALIZAÇÃO (2FN)

“Uma entidade está na segunda forma normal se cada atributo que não estiver na chave primária fornecer um fato que dependa da chave inteira.”

GAME_TEAM

<u>**matchID**</u>	<u>**gameset**</u>	**teamName**	**teamRegion**	baron	dragon	turrets

NORMALIZAÇÃO (2FN)

"Uma entidade está na segunda forma normal se cada atributo que não estiver na chave primária fornecer um fato que dependa da chave inteira."

PLAYER_GAME_STATS

matchID	**gameset**	**playerName**	teamName	Pick	Kills	Deaths	Assists	Farm	Gold	Damage	Tanking

NORMALIZAÇÃO (2FN)

"Uma entidade está na segunda forma normal se cada atributo que não estiver na chave primária fornecer um fato que dependa da chave inteira."

MATCH

<u>**matchID**</u>	<u>**gameset**</u>	matchType	matchDate	matchDuration	win

Match

matchID	matchType	matchDate

Game

<u>**matchID**</u>	**gameset**	duration	winner

APÓS 2FN

Ban

<u>matchID</u>	**<u>gameset</u>**	**teamName**	**banOrder**	champion

Game_Team

<u>matchID</u>	**<u>gameset</u>**	**teamName**	**teamRegion**	baron	dragon	turrets

Player_Game_Stats

<u>matchID</u>	**<u>gameset</u>**	**playerName**	teamName	Pick	Kills	Deaths	Assists	Farm	Gold	Damage	Tanking

Match

matchID	matchType	matchDate

Game

<u>matchID</u>	**gameset**	duration	winner

3º FORMA NORMAL

NORMALIZAÇÃO (3FN)

"Uma entidade está na terceira forma normal se cada atributo de chave não primária fornecer um fato que seja independente de outros atributos de chave não primária e dependa apenas da chave."

Ban

matchID	**gameset**	**teamName**	**banOrder**	champion

Game_Team

matchID	**gameset**	**teamName**	**teamRegion**	baron	dragon	turrets

Player_Game_Stats

matchID	**gameset**	**playerName**	teamName	Pick	Kills	Deaths	Assists	Farm	Gold	Damage	Tanking

Match

matchID	matchType	matchDate

Game

matchID	**gameset**	duration	winner

APÓS 3FN

Ban

matchID	**gameset**	**teamName**	**banOrder**	champion

Game_Team

matchID	**gameset**	**teamName**	**teamRegion**	baron	dragon	turrets

Player_Game_Stats

matchID	**gameset**	**playerName**	teamName	Pick	Kills	Deaths	Assists	Farm	Gold	Damage	Tanking

Match

matchID	matchType	matchDate

Game

matchID	**gameset**	duration	winner

MELHORIAS

MELHORIAS

sublinhado: chave estrangeira
amarelo: modificações
verde: adições

Champion

championID	championName

Ban

matchID	**gameset**	**teamID**	**banOrder**	championID

Game

matchID	**gameset**	duration	winnerID

Game_Team

matchID	**gameset**	**teamID**	baron	dragon	turrets

Team

teamID	teamName	teamRegion

Player_Game_Stats

matchID	**gameset**	**playerID**	teamID	championID	role	Kills	Deaths	Assists	Farm	Gold	Damage	Tanking

Player

playerID	playerName

Match

matchID	tournamentID	matchDate

Tournament

tournamentID	tournamentName

TABELAS FINAIS

TABELAS FINAIS

Champion

championID	championName

Ban

<u>matchID</u>	**<u>gameset</u>**	**<u>teamID</u>**	**banOrder**	<u>championID</u>

Game

<u>matchID</u>	**gameset**	duration	<u>winnerID</u>

Game_Team

<u>matchID</u>	**<u>gameset</u>**	**<u>teamID</u>**	baron	dragon	turrets

Team

teamID	teamName	teamRegion

Player_Game_Stats

<u>matchID</u>	**<u>gameset</u>**	**<u>playerID</u>**	<u>teamID</u>	<u>championID</u>	role	Kills	Deaths	Assists	Farm	Gold	Damage	Tanking

Player

playerID	playerName

Match

matchID	<u>tournamentID</u>	matchDate

Tournament

tournamentID	tournamentName

MODELO RELACIONAL

Champion(championID, championName)

Team(teamID, teamName, teamRegion)

Player(playerID, playerName)

Tournament(tournamentID, tournamentName)

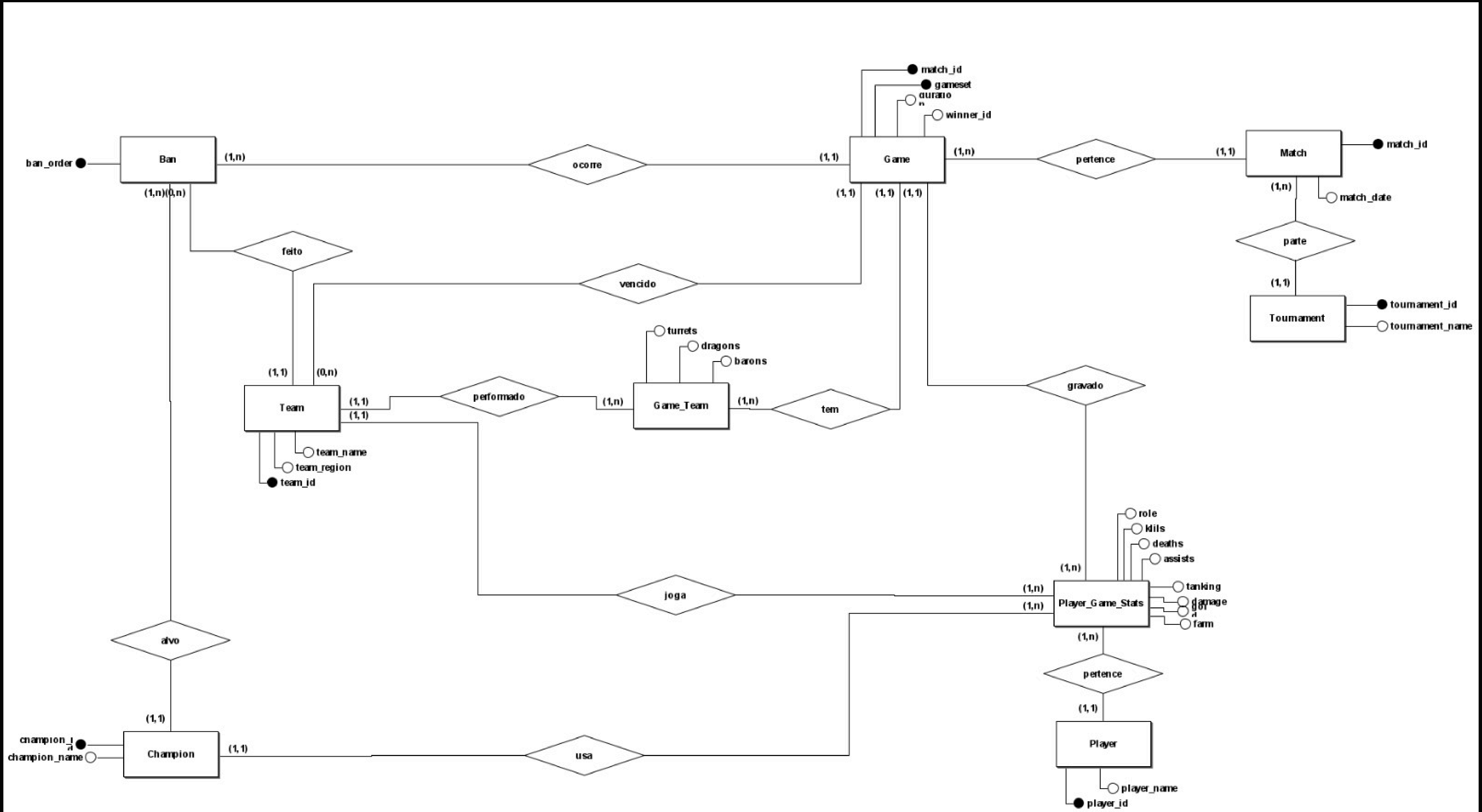
Match(matchID, tournamentID, matchDate)
tournamentID referencia Tournament

Game(matchID, gameset, duration, winnerID)
matchID referencia Match
winnerID referencia Team

Ban(matchID, gameset, teamID, banOrder, championID)
matchID referencia Match
gameset referencia Game
teamID referencia Team
championID referencia Champion

Game_Team(matchID, gameset, teamID, baron, dragon, turrets)
matchID referencia Match
gameset referencia Game
teamID referencia Team

Player_Game_Stats(matchID, gameset, playerID, teamID, championID, role, kills, deaths, assists, farm, gold, damage, tanking)
matchID referencia Match
gameset referencia Game
playerID referencia Player
teamID referencia Team
championID referencia Champion



Ban
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Game
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gameset
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winner_id
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Match
match_id
match_date
(1,n)
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(1,1)
Tournament
tournament_id
tournament_name
(1,n)(0,n)
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vencido
(1,1)
(0,n)
Team
team_name
team_region
team_id
(1,1)
(1,1)
performado
(1,n)
Game_Team
turrets
dragons
barons
(1,n)
tem
gravado
(1,n)
joga
(1,n)
(1,n)
Player_Game_Stats
role
kills
deaths
assists
tanking
damage
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Player
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Champion
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CÓDIGO

LÓGICA DO CÓDIGO

step_01_create_and_load_raw.py: Primeiramente lê o .csv, coleta cada coluna e o tipo de dado e cria uma tabela com todas as colunas. Após isso, insere todos os dados do .csv na tabela.
step_02_create_normalized_schema.py: Executa o script .SQL de criação das tabelas.
step_03_migrate_to_normalized: Roda sequencialmente vários scripts .SQL de inserção dos dados **raw_matches → tabelas normalizadas**.

scripts_sql: A inserção é feita seguindo a lógica
INSERT INTO tabela_normalizada1(
    SELECT (dados_raw_matches**) FROM (**raw_matches**)**
)
LEFT JOIN existing;

QUERIES

JUNÇÃO EXTERNA

Seleciona todos os nomes dos campeões e diz quantas vezes foram selecionados, ordenando pela quantidade, do maior para o menor.

```
SELECT
    c.champion_name,
    COUNT(jp.match_id) AS picks
FROM champion c
LEFT JOIN player_game_stats jp
    ON jp.champion_id = c.champion_id
GROUP BY c.champion_id, c.champion_name
ORDER BY picks DESC, c.champion_id;
```

AGREGAÇÃO

Seleciona as equipes com pelo menos 20 jogos e as médias de ouro das mesmas, retornando a região e o número total de partidas.

```
SELECT
    t.team_name,
    t.team_region,
    ROUND(AVG(pgs.gold), 2) AS media_gold,
    COUNT(DISTINCT CONCAT(pgs.match_id, '-', pgs.gameset)) AS jogos
FROM player_game_stats pgs
JOIN team t
    ON t.team_id = pgs.team_id
GROUP BY t.team_id, t.team_name, t.team_region
HAVING COUNT(DISTINCT CONCAT(pgs.match_id, '-', pgs.gameset)) >= 20
ORDER BY media_gold DESC;
```

JUNÇÃO 5 TABELAS

Mostra os jogadores com mais campeões diferentes e o seu número de partidas em ordem, com pelo menos 50 jogos.

```
SELECT
    p.player_name,
    COUNT(DISTINCT c.champion_id) AS
campeoes_diferentes,
    COUNT(*) AS partidas
FROM player_game_stats pgs
JOIN player p
    ON p.player_id = pgs.player_id
JOIN champion c
 ON c.champion_id = pgs.champion_id
JOIN game g
 ON g.match_id = pgs.match_id
 AND g.gameset = pgs.gameset
JOIN team t
 ON t.team_id = pgs.team_id
GROUP BY
 p.player_id,
 p.player_name
HAVING COUNT(*) >= 50
ORDER BY campeoes_diferentes DESC;
```

SUBCONSULTA

Seleciona os jogadores com mais de 50 partidas e o seu número de partidas.

```
SELECT p.player_name, COUNT(*) AS
qtd_partidas
FROM player p
JOIN player_game_stats pgs ON p.player_id
= pgs.player_id WHERE p.player_id IN
   (SELECT player_id
    FROM player_game_stats
    GROUP BY player_id
    HAVING COUNT(*) > 50)
GROUP BY p.player_id;
```

OPERADOR DE CONJUNTO

Seleciona todos os campeões com maior KDA médio por rota, tendo no mínimo 20 partidas.

```
(SELECT 'TOP' AS lane, c.champion_name AS champion,
ROUND((SUM(pgs.kills) + SUM(pgs.assists)) / NULLIF(SUM(pgs.deaths), 0), 2) AS kda
FROM player_game_stats pgs JOIN champion c ON c.champion_id =
pgs.champion_id WHERE pgs.role = 'TOP'
GROUP BY c.champion_id, c.champion_name
HAVING COUNT(*) >= 20
ORDER BY kda DESC LIMIT 1)
```

UNION ⟶ Vai fazer o mesmo para as próximas rotas, usando UNION para concatenar em uma saída.

OBRIGADO PELA ATEÇÃO!